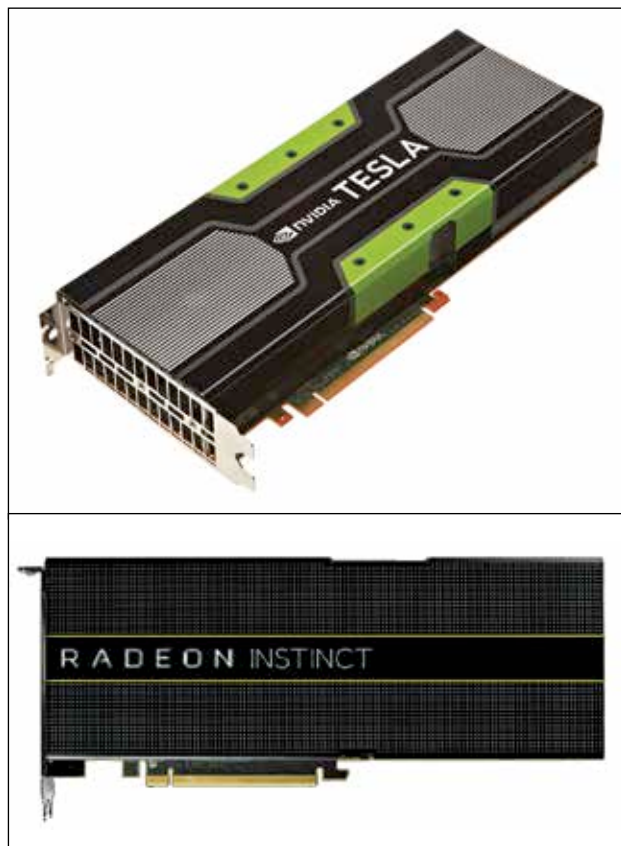


easier to get your hands on as more and more companies are releasing non-private data. Even those starting out can get their hands on voluminous data sets pertaining to, let's say Amazon Book Reviews or Dota2 Game matchups / results, Facebook comment volume data, etc.

Hardware is affordable

High Performance GPUs that can really accelerate the pace of Machine Learning like NVIDIA's Tesla and AMD's Radeon Instinct have been announced. While both companies have had



GPGPU is getting a massive push thanks to NVIDIA and AMD

their GPUs being used for OpenCL and proprietary GPGPU applications for a while, the move to smaller process nodes for their transistors have resulted in a tremendous rise of compute power. Needless to say, the basic solutions from either company have become really affordable. While industrial applications would still require spending ridiculous amounts of money to set up Machine Learning servers, it will be way easier for startups to dive into the deep end without the fear of sinking.

This widespread adoption will lead to several off-the-shelf solutions becoming a reality and thereafter, becoming even more affordable. Be it GPUs, services on the cloud or neural chipsets for the embedded market, the cost will go down.

Blockchain for all

Unless you've been living under a rock, you'd have heard about blockchain. It's a distributed database that maintains a steadily growing list of ordered records. They are inherently resistant to past records getting modified and have a high fault toler-



Banks are championing the development of Blockchain

ance. Conceptualised by Satoshi Nakamoto for the now infamous Bitcoin – a form of cryptocurrency – Blockchain is being readily considered by banks for issuance of bank issued legal currency. R3, a distributed database technology company that leads a consortium of over 70 financial institutions has been investing heavily in blockchain technology. R3 is seeking \$150 million from its member companies to fund the development of blockchain-based technologies for the banking sector, and they've already secured more than half of that amount. While a few top names have left the consortium – Goldman Sachs, Morgan Stanley, Santander and NAB – more have stepped in to take their place.

Aside from traditional applications, Blockchains are also being targeted at recording Smart Contracts. Simply put, Smart Contracts are a bunch of conditional statements that form the basis of a legal agreement between parties. The moment these conditions are met, the Smart Contract gets executed. Ethereum is a well-known platform that deals in Smart Contracts. And there are a lot many other such as Maidsafe, Ardor, Synereo, etc. which are all working towards creating the decentralised internet. By the time their efforts bear fruits, we should arrive at the end of 2017 and hopefully see Blockchain evolve.

IoT wreaks havoc

The Internet of Things is going to be huge, with over 6 billion devices that are connected to the internet – A figure that's expected to exponentially rise in the coming years to hit the 50 billion mark by 2020. However, there is no uniform security framework for IoT. All manufacturers implement their own proprietary solutions which has led to an extremely fragmented scenario. And in the midst of all this, we've had the very first massive IoT based botnet attack in October 2016. Now known as the Mirai botnet, this attack crippled the servers of companies like Netflix, NYTimes, Paypal and Twitter.

Cisco, AT&T, GE, Intel and IBM formed the Industrial Internet Consortium about two years back to focus on the IoT



Mirai Botnet was only the first